

These notes follow *Unitary Group Representations in Physics, Probability, and Number Theory* by George Mackey, *Group theory and physics* by Shlomo Sternberg, and Lucas Mason-Brown's notes from a representation theory course he taught (unpublished).

In these notes we describe a geometric procedure for inducing representations of subgroups, how to obtain representations of the “almost Abelian” semidirect products, and concluding with a brief description of the representation theory of the double cover of the identity component of the Poincaré group, minimizing details. Of course, if there are errors, please contact me.

1 Induction

1.1 Vector bundles coming from representations of subgroups

Let G be a locally compact second countable group. If X is a left G -set and G acts transitively on X , then X is isomorphic to the right coset space $G/\text{Stab}_G(x)$ for any $x \in X$ as left G -sets. We can promote this bijection of left G -sets to an isomorphism of left G -objects for suitable objects if we sprinkle in the appropriate adjectives in the correct places. We will focus on these so-called *homogeneous spaces* G/H (for H a closed subgroup) throughout these notes. In general a left G -space action naturally linearizes into a right action on function spaces: If G acts on the left of X , then G acts on the right of $\text{Fun}(X)$ by $(fg)(-) = f(g(-))$. We can also obtain a left action on function spaces: G acts on the left of $\text{Fun}(X)$ by $(gf)(-) = f(g^{-1}(-))$. By Fun we mean spaces of whatever looks like functions (i.e., functions, sections, cohomology classes, etc.). We will always linearize left actions on spaces to left actions on functions in what follows.

Let H be a closed subgroup of G and consider the principal H -bundle

$$\begin{array}{ccc} G & & g \\ \downarrow \pi & & \downarrow \\ G/H & & gH \end{array}$$

where H acts on the right of G by right multiplication and hence preserves the fibers of G . There is also a left action of G on G given by left multiplication, which commutes with the right action of H on G and makes G a G -equivariant bundle. The G -equivariance condition is that the actions of G on the total space and base space commute with π . Let (ρ, V) be a complex representation of H (which for now we treat as a vector space over $\mathbb{R}$). Consider the *associated vector bundle*

$$\begin{array}{ccc} G \times^H V & & (g, v) \\ \downarrow \pi & & \downarrow \\ G/H & & gH \end{array}$$

where $G \times^H V := (G \times V)/\sim$ and $(g, v) \sim (gh, \rho(h^{-1})v)$ for all $h \in H$ (equivalently $(gh, v) \sim (g, \rho(h)v)$ for

all $h \in H$). By definition of the relation $\sim$, the fibers of $G \times^H V$ are copies of V . This bundle is moreover a G -equivariant vector bundle, with G acting on the left of $G \times^H V$ by $g'(g, v) = (g'g, v)$.

Global sections of $G \times^H V$ are maps $s: G/H \rightarrow G \times^H V$ satisfying $\pi \circ s = \text{id}_{G/H}$, which form a vector space $\Gamma(G/H, G \times^H V)$. Since G acts on G/H on the left we obtain a left action on $\Gamma(G/H, G \times^H V)$ by $(gs)(-) = gs(g^{-1}(-))$. Observe that the G -equivariance of $G \times^H V$ is necessary for this action to be defined; that is, in order for $\pi \circ (gs) = \text{id}_{G/H}$. Moreover, the calculations

$$\begin{aligned} s(gH) &= (g, f(g)) = (gh, \rho(h^{-1})f(g)) \\ s(ghH) &= (gh, f(gh)) \end{aligned}$$

imply that $f(gh) = \rho(h^{-1})f(g)$ for all $h \in H$, so we can identify the global sections $\Gamma(G/H, G \times^H V)$ of $G \times^H V$ with the function space

$$\{f: G \rightarrow V \mid f(gh) = \rho(h^{-1})f(g) \text{ for all } h \in H\}.$$

The left action of G on $\Gamma(G/H, G \times^H V)$ is transported to the left action $(gf)(-) = f(g^{-1}(-))$ on the above function space. This function space is usually denoted by $\text{Ind}_H^G(\rho)$ or $\text{Ind}_H^G(V)$, and is called an induced representation of G (of the representation (ρ, V)).

When G is a real Lie group, there is an equivalence of the categories of finite-rank G -equivariant vector bundles on G/H and finite-dimensional smooth representations of H ; one functor is the associated bundle construction and the inverse is given by taking the fiber at a point.

1.2 Densities on smooth manifolds

Let $s > 0$. On a real vector space V of dimension n , s -densities are maps of sets $\mu: V^n \rightarrow \mathbb{R}$ such that for any $A \in \text{End}(V)$,

$$\mu(Av_1, \dots, Av_n) = |\det A|^s \mu(v_1, \dots, v_n).$$

Denote by $D^s(V)$ the set of all s -densities on V . It is the case that $D^s(V)$ is a one-dimensional real vector space under pointwise addition and scalar multiplication, and we may identify $D^s(V)$ with $|\bigwedge^n V^*|^s$, where the $|\cdot|^s$ is a suggestive notation.

Every s -density on V is either positive, negative, or the zero density. Indeed, choose a basis $\{e_1, \dots, e_n\}$ of V so that for any $v_1, \dots, v_n \in V$, there exists $A \in \text{End}(V)$ for which $Ae_i = v_i$. Then the map $D^s(V) \rightarrow \mathbb{R}$ given by $\mu \mapsto \mu(e_1, \dots, e_n)$ is an injection since an s -density μ is the zero density if and only if $\mu(e_1, \dots, e_n) = 0$, since

$$\mu(v_1, \dots, v_n) = |\det A|^s \mu(e_1, \dots, e_n).$$

It remains to exhibit a nonzero density: for this consider the map

$$(v_1, \dots, v_n) \mapsto |\det A|^s$$

for $A \in \text{End}(V)$ as before, and this is a density since the determinant is multiplicative. For any s -density μ , the earlier calculation $\mu(v_1, \dots, v_n) = |\det A|^s \mu(e_1, \dots, e_n)$ implies that the sign of μ is the sign of $\mu(e_1, \dots, e_n)$.

Let X be a smooth real manifold of dimension n . The frame bundle $F(TX)$ of the tangent bundle, called the tangent frame bundle, is the bundle over X of all ordered bases of the tangent spaces at points of X ; that is, the fiber over $x \in X$ is the set of ordered bases of $T_x X$. By an ordered basis of a real vector space V we mean a vector space isomorphism $\beta: \mathbb{R}^n \rightarrow V$. The tangent frame bundle is a principal $\text{GL}_n(\mathbb{R})$ -bundle; the right action on the fiber over p is given by precomposing an isomorphism $\mathbb{R}^n \rightarrow T_x X$ with multiplication by an element of $\text{GL}_n(\mathbb{R})$.

Consider the left action of $\text{GL}_n(\mathbb{R})$ on $D^s(\mathbb{R}^n)$ by $A\mu = \mu(A^{-1}-, \dots, A^{-1}-) = |\det A|^{-s} \mu$ (this is some kind of left regular action if we identify $(\mathbb{R}^n)^n$ with $\mathbb{R}^{n \times n}$ by concatenation). This defines a one-dimensional representation of $\text{GL}_n(\mathbb{R})$, so we can form the associated bundle

$$D_X^s := F(TX) \times^{\text{GL}_n(\mathbb{R})} D^s(\mathbb{R}^n) = F(TX) \times D^s(\mathbb{R}^n) / \sim,$$

where $(\beta, \mu) \sim (\beta A, A^{-1}\mu) = (\beta \circ (A \cdot -), |\det A|^s \mu)$. We call this bundle the bundle of s -densities on X . The map $(\beta, \mu) \mapsto \mu \circ (\beta^{-1})^{\oplus n}$ defines an isomorphism of the fiber over $x \in X$ of D_X^s with $D^s(T_x X)$, so we can think of D_X^s as $\coprod_{x \in X} D^s(T_x X)$; this also shows that D_X^s is a line bundle. Note that D_X^s is covariant (maybe we shouldn't call it a functor?), taking a local diffeomorphism $f: X \rightarrow Y$ to the map $(\beta, \mu) \mapsto (df \circ \beta, \mu)$, which on fibers is just pushforward along df .

Global sections of D_X^s are called s -densities on X . These s -densities pull back along maps $X \rightarrow Y$ in a similar way to differential forms, and pulling back preserves the sign of the s -density. If X is orientable and $\omega \in \Omega^n(X)$ is a top form on X , then $|\omega|^s$ is a smooth s -density on X (and every s -density on X appears this way!). In general, for any X , the bundle of s -densities on X is nontrivial. We can construct one by using a partition of unity $\{c_j\}_j$ subordinate to an atlas $\{U_j\}_j$ of X , choosing positive s -densities δ_j on each U_j (e.g. by pulling back a positive s -density on the copy of $\mathbb{R}^n$ each U_j is homeomorphic to), and forming the sum $\sum_j c_j \delta_j$, which is the desired nontrivial s -density on X .

The compactly supported continuous 1-densities on X may be integrated, by passing to local coordinates and integrating over $\mathbb{R}^n$ using the Lebesgue integral. This is because the continuous 1-densities on $\mathbb{R}^n$ are of the form $f|dx_1 \wedge \dots \wedge dx_n|$ for $f \in C(\mathbb{R}^n)$ a real-valued continuous function on $\mathbb{R}^n$, and the change of coordinates formula for these densities matches the corresponding change of coordinates for the Lebesgue measure on the real line. Everything works as expected, but notably if $f: X \rightarrow Y$ is a diffeomorphism and $s \in \Gamma_c(Y, D^1(Y))$, then

$$\int_X f^* s = \int_Y s.$$

Given an s -density and a t -density on X , we may multiply them together to obtain an $(s+t)$ -density on X . In other words, multiplication defines a linear map $\Gamma(X, D_X^s) \otimes_{\mathbb{R}} \Gamma(X, D_X^t) \rightarrow \Gamma(X, D_X^{s+t})$. Therefore we can form an inner product on the vector space $\Gamma_c(X, D_X^{1/2})$ of compactly supported smooth half-densities on X ,

$$\langle -, - \rangle: \Gamma_c(X, D_X^{1/2}) \times \Gamma_c(X, D_X^{1/2}) \rightarrow \mathbb{R},$$

given by multiplication and then integration.

1.3 Unitary induction for real Lie groups

Let G be a real Lie group with H a closed subgroup, and let $\mathfrak{g} = \text{Lie}(G)$, $\mathfrak{h} = \text{Lie}(H)$ be the corresponding Lie algebras. We are interested in the homogeneous space $X = G/H$.

Since G acts on X by diffeomorphisms, it follows that D_X^s is a G -equivariant line bundle. The fiber over $1_G H$ is isomorphic to $D^s(T_{1_G H}(G/H)) = D^s(\mathfrak{g}/\mathfrak{h}) \cong |\wedge^{\text{top}}(\mathfrak{g}/\mathfrak{h})^*|^s$, and the action of H on this one-dimensional real vector space is given by the character $h \mapsto |\det(\text{Ad}(h)^{-1}|_{\mathfrak{g}/\mathfrak{h}})|^s$ (a fun fact).

Let (ρ, V) be a unitary representation of H with H -invariant Hermitian inner product $\langle -, - \rangle_V$. Then form the associated bundle $G \times^H V$ on X and take the tensor product with the bundle of half-densities $D^{1/2}(X)$ on X . The compactly supported continuous sections of $(G \times^H V) \otimes D^{1/2}(X)$ form an inner product space, via the Hermitian pairing

$$\langle -, - \rangle : \Gamma_c(X, (G \times^H V) \otimes D^{1/2}(X)) \otimes_{\mathbb{R}} \Gamma_c(X, (G \times^H V) \otimes D^{1/2}(X)) \rightarrow \Gamma_c(X, D^1(X)) \rightarrow \mathbb{C},$$

which explicitly looks like

$$\langle f|dx|^{1/2}, g|dx|^{1/2} \rangle = \int_X \langle f, g \rangle_V |dx|,$$

and this pairing is G -invariant. The Hilbert space completion of $\Gamma_c(X, (G \times^H V) \otimes D^{1/2}(X))$ is the induced representation of V to a representation of G ; it is some kind of L^2 space. We can identify this space with the space of functions

$$\{f : G \rightarrow V \mid f(gh) = \rho(h^{-1})f(g) \text{ for all } h \in H; \|f\|_{L^2}^2 < \infty\},$$

where $\|f\|_{L^2}^2 = \int_{G/H} \langle f, f \rangle_V |dx|$ (using $|dx|$ to denote the unsigned “measure” we get on G/H which is left G -invariant since I don’t know a better notation). This Hilbert space of functions is denoted by $\text{Ind}_H^G(\rho)$ or $\text{Ind}_H^G(V)$.

Unitary induction can be done in the case when G is merely a locally compact second countable group, but since G/H need not admit a left G -invariant Haar measure, we have to use something called a “Hilbert bundle” to get around this issue. This procedure and more (i.e. discussion of representations of semidirect products) is explained in Unitary Group Representations in *Physics, Probability, and Number Theory* by Mackey in sections 7 through 11.

2 Semidirect products

Let G be a real Lie group which is the semidirect product of closed subgroups N and H ; that is, $G = N \rtimes_{\phi} H$ where $\phi: H \rightarrow \text{Aut}(N)$ is smooth. As a set, $G = N \rtimes_{\phi} H = N \times H$ with group law

$$(n_1, h_1)(n_2, h_2) = (n_1\phi(h_1)n_2, h_1h_2).$$

We now restrict to the case where N is Abelian, and we call groups G appearing as $N \rtimes_{\phi} H$ “almost Abelian”. To find a (unitary, irreducible) representation of G , we will carry out a prescribed sequence of inductions of representations, which physicists call “the method of little groups”. This method is attributed to Wigner and also Mackey. The recipe is as follows:

Step 1. Since N is Abelian, consider the dual group $\widehat{N} = \text{Hom}(N, S^1)$ which parameterize the irreducible (one-dimensional) representations of N . Then both H and N act on $\widehat{N}$ by

$$\begin{aligned} h\chi &= \chi(h(-)h^{-1}) \quad \text{and} \\ n\chi &= \chi(n(-)n^{-1}) = \chi. \end{aligned}$$

Note that N acts trivially on $\widehat{N}$. By precomposing characters $\chi: N \rightarrow S^1$ with the projection to the first component map $\pi_1: N \rtimes_{\phi} H \rightarrow N$, we obtain an action of $G = N \rtimes_{\phi} H$ acts on $\widehat{N}$ by

$$(n, h)\chi = \chi(\pi_1((n, h)(-, 1_H)(n, h)^{-1})) = \chi(\pi_1((1_N, h)(-, 1_H)(1_N, h)^{-1})).$$

Step 2. Calculate the orbits of the action of G on $\widehat{N}$, so that

$$\widehat{N} = \bigsqcup_{i \in I} G\chi_i \quad \text{and} \quad G\chi_i \cong G/\text{Stab}_G(\chi_i).$$

Since N acts trivially on $\widehat{N}$, $\text{Stab}_G(\chi_i) \cong N \rtimes_{\phi} L_i$ for some closed subgroup L_i of H . We call L_i a “little group”.

Step 3. Take a representation (ρ, V) of L_i and induce it to a representation of $N \rtimes_{\phi} L_i = \text{Stab}_G(\chi_i)$; explicitly the action looks like

$$(n, h)v = \chi_i(n)\rho(h)v.$$

Denote this representation by $\text{Ind}_{L_i}^{\text{Stab}_G(\chi_i)}(V)$.

Step 4. Finally induce the representation $\text{Ind}_{L_i}^{\text{Stab}_G(\chi_i)}(V)$ of $\text{Stab}_G(\chi_i)$ to a representation of G ; by doing this we obtain the function space $W = \text{Ind}_{\text{Stab}_G(\chi_i)}^G(\text{Ind}_{L_i}^{\text{Stab}_G(\chi_i)}(V))$. We obtain a unitary representation of G if (ρ, V) was a unitary representation of L_i , but it's not clear that we obtain an irreducible representation of G if (ρ, V) was an irreducible representation of L_i .

It is a theorem that the representation W is irreducible if and only if (ρ, V) is irreducible. But we can see this sort of intuitively by seeing that the induction steps roughly just twist ρ by a character, and then smears the

intermediate representation over the orbit $G\chi_i$ (note that G acts transitively on $G\chi_i$) to form the bundle whose sections are W . These steps seem to be reversible, and it might be believable that the steps which reverse the recipe above preserve reducibility. A better explanation is in this StackExchange answer, for finite groups.

3 The Lorentz group

3.1 Definitions

The space $\mathbb{R}^4 = \mathbb{R}^{1+3}$ is called spacetime, and has metric given by any matrix g with signature $\text{diag}(+, -, -, -)$. The orthogonal group preserving this metric is

$$\text{O}(1, 3) = \{A \in \text{Mat}_{4 \times 4}(\mathbb{R}) \mid A^T g A = g\} \subseteq \text{O}(4).$$

The group $\text{O}(1, 3)$ is called the homogeneous Lorentz group. By viewing $\mathbb{R}^{1+3}$ as an affine space, the isometries of $\mathbb{R}^{1+3}$ are captured by the group $\mathbb{R}^{1+3} \rtimes \text{O}(1, 3)$ (where the action of $\text{O}(1, 3)$ on $\mathbb{R}^{1+3}$ is the usual one, so we suppress writing down the homomorphism $\text{O}(1, 3) \rightarrow \text{Aut}(\mathbb{R}^{1+3})$). The group $\mathbb{R}^{1+3} \rtimes \text{O}(1, 3)$ is called the inhomogeneous Lorentz group, also called the Poincaré group. The Lorentz group is the group of changes of coordinates of spacetime which preserve the speed of light, so it and the Poincaré group are studied in special relativity.

The homogeneous Lorentz group is a six-dimensional real Lie group which is not compact and not connected. The four connected components are each not simply connected (with fundamental group $\mathbb{Z}/2\mathbb{Z}$), and each component is determined by the values of the homomorphisms

$$\text{O}(1, 3) \xrightarrow{\det} \{\pm 1\} \quad \text{and} \quad \text{O}(1, 3) \xrightarrow{\text{sgn}(-_{00})} \{\pm 1\}$$

where $\text{sgn}(A_{00})$ is the sign of A_{00} . The proper (or special) Lorentz group is $\ker(\det) =: \text{SO}(1, 3)$, and preserves the orientation of spacetime. The orthochronous Lorentz group is $\ker(\text{sgn}(-_{00})) =: O^\uparrow(1, 3)$, and preserves the orientation of time. The connected component of the identity

$$\text{SO}^\uparrow(1, 3) := \text{SO}(1, 3) \cap O^\uparrow(1, 3)$$

is the proper orthochronous (or also called restricted) Lorentz group.

The double cover (universal cover) of the proper orthochronous Lorentz group $\text{SO}^\uparrow(1, 3)$ is $\text{SL}_2(\mathbb{C})$: Consider the space of 2×2 Hermitian matrices, which is the $\mathbb{R}$ -span of the identity and Pauli matrices

$$\begin{pmatrix} 1 & \\ & 1 \end{pmatrix}, \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}, \begin{pmatrix} & -i \\ i & \end{pmatrix}, \quad \text{and} \quad \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}.$$

Thus any Hermitian matrix is of the form

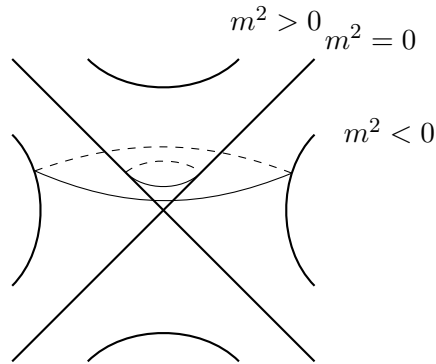
$$\begin{pmatrix} t + z & x - iy \\ x + iy & t - z \end{pmatrix}.$$

Crucially, the determinant of this matrix is $t^2 - x^2 - y^2 - z^2$, so we may identify $\mathbb{R}^{1+3}$ with the space of Hermitian matrices. The group $\mathrm{SL}_2(\mathbb{C})$ acts on the space of Hermitian matrices by $H \mapsto AHA^*$ for H a Hermitian matrix and $A \in \mathrm{SL}_2(\mathbb{C})$. It follows that $\mathrm{SL}_2(\mathbb{C})$ acts on $\mathbb{R}^{1+3}$, and we obtain a representation $\kappa: \mathrm{SL}_2(\mathbb{C}) \rightarrow \mathrm{GL}(\mathbb{R}^{1+3})$. It can be shown that $\mathrm{im}(\kappa) \subseteq \mathrm{O}(1, 3)$, $\ker(\kappa) = \{\pm I\}$, and that κ is a local isomorphism. Since $\mathrm{SL}_2(\mathbb{C})$ is connected, $\mathrm{im}(\kappa)$ is $\mathrm{SO}^\uparrow(1, 3)$.

3.2 Representation theory

We follow Wigner's approach. To avoid dealing with possibly projective representations of the Poincaré group (as it is not simply connected), we pass to the double cover of its identity component, $\mathbb{R}^{1+3} \rtimes_{\kappa} \mathrm{SL}_2(\mathbb{R})$. As a consequence, we may obtain more representations than we originally intended to, and in some sense we are missing some representations as well, since we are not able to obtain representations of the whole Poincaré group this way (we do not incorporate space parity inversion $P = \mathrm{diag}(1, -1, -1, -1)$ or time reversal $T = \mathrm{diag}(-1, 1, 1, 1)$ this way).

Following the recipe from the previous section, in order to determine the representation theory of $\mathbb{R}^{1+3} \rtimes_{\kappa} \mathrm{SL}_2(\mathbb{R})$, we first calculate the orbits of the action of $\mathrm{SL}_2(\mathbb{C})$ on $\widehat{\mathbb{R}^{1+3}}$, which we identify with $\mathbb{R}^{1+3}$. We will also continue to identify spacetime with the space of Hermitian matrices. If H is a Hermitian matrix, the action of $\mathrm{SL}_2(\mathbb{C})$ preserves the determinant. If we denote the determinant of a generic Hermitian matrix by m^2 (so from an earlier description, this means $m^2 = t^2 - x^2 - y^2 - z^2$), then the orbits of the action of $\mathrm{SL}_2(\mathbb{C})$ on $\mathbb{R}^{1+3}$ are contained in the following level sets of m^2 :



The origin is an orbit. Thus for the level sets where $m^2 > 0$ and $m^2 = 0$ the level sets each have two sheets corresponding to whether t is positive or negative, and each sheet is an orbit. For $m^2 = 0$ the sheets are suggestively called the forward and backwards light cones. The level sets for $m^2 < 0$ are each connected and hence are orbits. What remains now is a lot of calculations according to the recipe above, so we summarize them in the table below.

orbit	representing point χ_i	Little group L_i
$m^2 > 0, t > 0$	$\begin{pmatrix} m & \\ & m \end{pmatrix}$	$\mathrm{SU}(2)$
$m^2 > 0, t < 0$	$\begin{pmatrix} -m & \\ & -m \end{pmatrix}$	$\mathrm{SU}(2)$
$m^2 = 0, t > 0$	$\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$	$\mathbb{R}^2 \rtimes_{2\theta} \mathrm{SO}(2)$
$m^2 = 0, t < 0$	$\begin{pmatrix} -2 & 0 \\ 0 & 0 \end{pmatrix}$	$\mathbb{R}^2 \rtimes_{2\theta} \mathrm{SO}(2)$
$m^2 = 0, t = 0$	$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$	$\mathrm{SL}_2(\mathbb{C})$
$m^2 < 0$	$\begin{pmatrix} & m i \\ - m i & \end{pmatrix}$	$\mathrm{SL}_2(\mathbb{R})$

Without getting into the details (since these are well documented elsewhere), the irreps of $\mathrm{SU}(2)$ are parameterized by half-integers $s = k/2$ for $k \geq 0$. They are unitary and finite-dimensional. The irreps of $\mathbb{R}^2 \rtimes_{2\theta} \mathrm{SO}(2)$ (where the 2θ just indicates that $\mathrm{SO}(2)$ acts on $\mathbb{R}^2$ by rotation through twice the specified angle, but this is not so important) may be obtained by the same recipe as before, and of these, the “physically relevant representations” are parameterized by half-integers $s = k/2$ for $k \in \mathbb{Z}$. They are unitary and one-dimensional. From $\mathrm{SL}_2(\mathbb{C})$ we can obtain finite-dimensional representations parameterized by half-integers $s = k/2$ for $k \geq 0$ which are not unitary (aside from the one yielding the trivial representation). We may also obtain unitary infinite-dimensional representations from the so-called principal series, parameterized by the line $i\mathbb{R}$ and the choice of a sign 0, 1, or from the complementary series parameterized by an open interval, say $0 < t < 2$. The representation theory of $\mathrm{SL}_2(\mathbb{R})$ is similar.

The variable names m and s are suggestive. The physically relevant representations should correspond to particles with mass m and spin s . But this is beyond my understanding, since I do not know any physics.